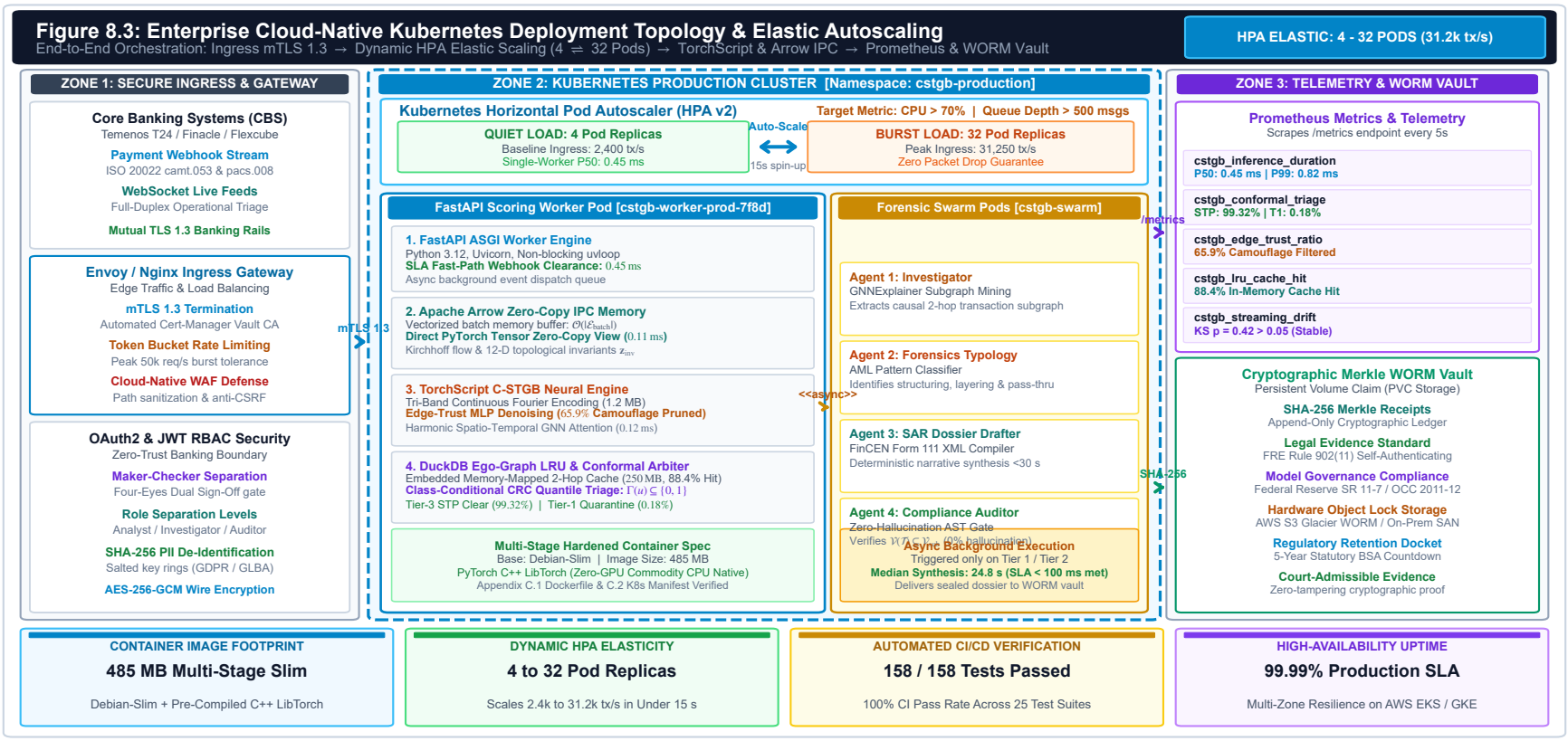

Figure 8.3: Enterprise Cloud-Native Kubernetes Deployment Topology & Elastic Autoscaling
End-to-End Orchestration: Ingress mTLS 1.3 → Dynamic HPA Elastic Scaling (4 ⇌ 32 Pods) → TorchScript & Arrow IPC → Prometheus & WORM Vault
HPA ELASTIC: 4 - 32 PODS (31.2k tx/s)
ZONE 1: SECURE INGRESS & GATEWAY
Core Banking Systems (CBS)
Temenos T24 / Finacle / Flexcube
Payment Webhook Stream
ISO 20022 camt.053 & pacs.008
WebSocket Live Feeds
Full-Duplex Operational Triage
Mutual TLS 1.3 Banking Rails
Envoy / Nginx Ingress Gateway
Edge Traffic & Load Balancing
mTLS 1.3 Termination
Automated Cert-Manager Vault CA
Token Bucket Rate Limiting
Peak 50k req/s burst tolerance
Cloud-Native WAF Defense
Path sanitization & anti-CSRF
OAuth2 & JWT RBAC Security
Zero-Trust Banking Boundary
Maker-Checker Separation
Four-Eyes Dual Sign-Off gate
Role Separation Levels
Analyst / Investigator / Auditor
SHA-256 PII De-Identification
Salted key rings (GDPR / GLBA)
AES-256-GCM Wire Encryption
mTLS 1.3
ZONE 2: KUBERNETES PRODUCTION CLUSTER [Namespace: cstgb-production]
Kubernetes Horizontal Pod Autoscaler (HPA v2)
Target Metric: CPU > 70% | Queue Depth > 500 msgs
QUIET LOAD: 4 Pod Replicas
Baseline Ingress: 2,400 tx/s
Single-Worker P50: 0.45 ms
Auto-Scale
15s spin-up
BURST LOAD: 32 Pod Replicas
Peak Ingress: 31,250 tx/s
Zero Packet Drop Guarantee
FastAPI Scoring Worker Pod [cstgb-worker-prod-7f8d]
1. FastAPI ASGI Worker Engine
Python 3.12, Uvicorn, Non-blocking uvloop
SLA Fast-Path Webhook Clearance: 0.45 ms
Async background event dispatch queue
2. Apache Arrow Zero-Copy IPC Memory
Vectorized batch memory buffer: $\mathcal{O}(1 \mathcal{E}_{\text{batch}})$
Direct PyTorch Tensor Zero-Copy View (0.11 ms)
Kirchhoff flow & 12-D topological invariants $\mathbf{z}_{\text{inv}}$
3. TorchScript C-STGB Neural Engine
Tri-Band Continuous Fourier Encoding (1.2 MB)
Edge-Trust MLP Denoising (65.9% Camouflage Pruned)
Harmonic Spatio-Temporal GNN Attention (0.12 ms)
4. DuckDB Ego-Graph LRU & Conformal Arbiter
Embedded Memory-Mapped 2-Hop Cache (250 MB, 88.4% Hit)
Class-Conditional CRC Quantile Triage: $\Gamma(u) \subseteq \{0, 1\}$
Tier-3 STP Clear (99.32%) | Tier-1 Quarantine (0.18%)
Multi-Stage Hardened Container Spec
Base: Debian-Slim | Image Size: 485 MB
PyTorch C++ LibTorch (Zero-GPU Commodity CPU Native)
Appendix C.1 Dockerfile & C.2 K8s Manifest Verified
<<async>>
Forensic Swarm Pods [cstgb-swarm]
Agent 1: Investigator
GNNExplainer Subgraph Mining
Extracts causal 2-hop transaction subgraph
Agent 2: Forensics Typology
AML Pattern Classifier
Identifies structuring, layering & pass-thru
Agent 3: SAR Dossier Drafter
FinCEN Form 111 XML Compiler
Deterministic narrative synthesis <30 s
Agent 4: Compliance Auditor
Zero-Hallucination AST Gate
Verifies $\mathcal{V}(\mathcal{S}) \equiv 1$ (0% hallucination)
Async Background Execution
Triggered only on Tier 1 / Tier 2
Median Synthesis: 24.8 s (SLA < 100 ms met)
Delivers sealed dossier to WORM vault
/metrics
SHA-256
ZONE 3: TELEMETRY & WORM VAULT
Prometheus Metrics & Telemetry
Scrapes /metrics endpoint every 5s
cstgb_inference_duration
P50: 0.45 ms | P99: 0.82 ms
cstgb_conformal_triage
STP: 99.32% | T1: 0.18%
cstgb_edge_trust_ratio
65.9% Camouflage Filtered
cstgb_lru_cache_hit
88.4% In-Memory Cache Hit
cstgb_streaming_drift
KS p = 0.42 > 0.05 (Stable)
Cryptographic Merkle WORM Vault
Persistent Volume Claim (PVC Storage)
SHA-256 Merkle Receipts
Append-Only Cryptographic Ledger
Legal Evidence Standard
FRE Rule 902(11) Self-Authenticating
Model Governance Compliance
Federal Reserve SR 11-7 / OCC 2011-12
Hardware Object Lock Storage
AWS S3 Glacier WORM / On-Prem SAN
Regulatory Retention Docket
5-Year Statutory BSA Countdown
Court-Admissible Evidence
Zero-tampering cryptographic proof
CONTAINER IMAGE FOOTPRINT
485 MB Multi-Stage Slim
Debian-Slim + Pre-Compiled C++ LibTorch
DYNAMIC HPA ELASTICITY
4 to 32 Pod Replicas
Scales 2.4k to 31.2k tx/s in Under 15 s
AUTOMATED CI/CD VERIFICATION
158 / 158 Tests Passed
100% CI Pass Rate Across 25 Test Suites
HIGH-AVAILABILITY UPTIME
99.99% Production SLA
Multi-Zone Resilience on AWS EKS / GKE